Gradient ascent for logistic regression

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1 Likelyhood function

$$\ell(w) = \prod_{i=0}^{n-1} P(y_i|x_i, w),$$

2 Log likelyhood function

$$\ell\ell(w) = \sum_{i=0}^{n-1} ((\mathbb{1}[y_i = +1] - 1)w^\top h(x_i) - \ln(1 + \mathrm{e}^{-w^\top h(x_i)}))$$

3 Derivative of log of the likelyhood funciton

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w))$$